DVR & Dr. HS MIC COLLEGE OF TECHNOLOGY

(Autonomous Institution | Approved by AICTE, Accredited by NBA & NAAC A+)

Kanchikacherla, NTR District, Andhra Pradesh - 521180

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING / CSE

B.Tech V Semester | Regulation: MIC23 | Subject Code: 23AM5T03 / 23CS4T01

OPERATING SYSTEMS

MASTER QUESTION BANK (UNITS I TO V)

OFFICIAL REPOSITORY: 2-MARK, 5-MARK, 10-MARK & NUMERICAL PROBLEMS

AUTHENTIC EXAM SOURCE VERIFICATION

Every question in this repository has been strictly compiled from Official Autonomous Model Question Papers, Mid-1 / Mid-2 Question Banks, and End Semester Examination Papers under Autonomous Regulation MIC23.

QUESTION BANK CONTENT BREAKDOWN & MARK DISTRIBUTION

Unit	Unit Title	2-Mark	5-Mark	10-Mark	Problems	Total
Unit I	OS Overview & System Structures	14	16	2	-	32
Unit II	Processes, Threads & CPU Scheduling	13	12	2	5	32
Unit III	Synchronization & Deadlocks	16	14	2	2	34
Unit IV	Memory Management & Mass Storage	10	12	2	7	31
Unit V	File Systems & Protection	7	13	2	-	22
TOTAL	**COMPLETE SYLLABUS COVERAGE**	**60**	**67**	**10**	**14**	**151 Questions**

Note: This document contains only questions without answers. For full step-by-step answers, refer to the Unit-wise Complete Notes.

MIC23 AUTONOMOUS QUESTION BANK REPOSITORY

UNIT - I : OPERATING SYSTEMS OVERVIEW & SYSTEM STRUCTURES

SOURCE VERIFICATION: Questions meticulously extracted from Official University Model Papers, Mid-1 / Mid-2 Question Banks, and Autonomous Semester End Examinations.

Regulation: MIC23 | Course Code: 23AM5T03 / 23CS4T01 | B.Tech V Semester (AI&ML / CSE)

UNIT - I - SECTION A: 2-MARK SHORT ANSWER QUESTIONS [2 MARKS EACH]

Q1. Define batch processing system.
[2M] • *Source: University Exam 2026*

Q2. What are layered operating systems?
[2M] • *Source: University Exam 2026*

Q3. What is an Operating System?
[2M] • *Source: Model Paper-I / Mid-1 Imp*

Q4. What are system calls in operating systems? / Define system calls.
[2M] • *Source: Model Paper-I / Mid-1 Imp*

Q5. List any two functions of an operating system.
[2M] • *Source: Model Paper-II*

Q6. What is the difference between open-source and proprietary operating systems?
[2M] • *Source: Model Paper-II*

Q7. Define Kernel.
[2M] • *Source: Mid-1 Imp*

Q8. What is multiprogramming?
[2M] • *Source: Mid-1 Imp*

Q9. How Dual-Mode Operation works?
[2M] • *Source: Mid-1 Imp*

Q10. What are the objectives of operating systems?
[2M] • *Source: Mid-1 Imp*

Q11. What is the use of fork and exec system calls?
[2M] • *Source: Mid-1 Imp*

Q12. Define open source operating system.
[2M] • *Source: Mid-1 Imp*

Q13. List the services provided by an Operating System.
[2M] • *Source: Mid-1 Imp*

Q14. What is the purpose of system programs?
[2M] • *Source: Mid-1 Imp*

UNIT - I: OPERATING SYSTEMS OVERVIEW & SYSTEM STRUCTURES

SECTION B: 5-MARK LONG ANSWER QUESTIONS [5 MARKS EACH]

Q1. Explain in detail the functions of an operating system.
[5M] • *Source: University Exam 2026 / Model Paper-I / Mid-1 Imp*

Q2. Describe the types of system calls with examples.
[5M] • *Source: University Exam 2026 / Model Paper-II / Mid-1 Imp*

Q3. Discuss the importance of debugging in operating system development.
[5M] • *Source: University Exam 2026 / Mid-1 Imp*

Q4. Explain in detail the structure of an operating system.
[5M] • *Source: University Exam 2026 / Mid-1 Imp*

Q5. Explain the difference between system programs and application programs.
[5M] • *Source: Model Paper-I*

Q6. Write short notes on open-source operating systems.
[5M] • *Source: Model Paper-I*

Q7. Explain layered and microkernel structures of operating systems with neat sketch.
[5M] • *Source: Model Paper-I*

Q8. Define an operating system and explain its main objectives.
[5M] • *Source: Model Paper-II*

Q9. Explain the steps involved in building and booting an operating system.
[5M] • *Source: Model Paper-II*

Q10. What are the main services provided by an operating system?
[5M] • *Source: Model Paper-II / Mid-1 Imp*

Q11. Explain about the dual mode operation in OS with a neat block diagram.
[5M] • *Source: Mid-1 Imp*

Q12. What is an operating system? Explain Computing Environments.
[5M] • *Source: Mid-1 Imp*

Q13. Write the differences between monolithic kernel and microkernel.
[5M] • *Source: Mid-1 Imp*

Q14. Explain briefly User and Operating System Interface (CLI, GUI, Touch).
[5M] • *Source: Mid-1 Imp*

Q15. Write short notes on Operating System Design and Implementation.
[5M] • *Source: Mid-1 Imp*

Q16. Explain System Programs in detail.
[5M] • *Source: Mid-1 Imp*

UNIT - I: OPERATING SYSTEMS OVERVIEW & SYSTEM STRUCTURES

SECTION C: 10-MARK COMPREHENSIVE ESSAY QUESTIONS (LAQs) [10 MARKS EACH]

Q1. Explain various operating system structures with neat sketches. Compare monolithic and microkernel architectures in detail.

[10M] • *Source: Model Paper-I / University Exam 2026 / Mid-1 Imp*

Q2. Explain Dual-Mode Operation and Hardware Protection in Operating Systems with neat sketches. Detail how timer interrupts prevent CPU monopoly.

[10M] • *Source: Mid-1 Imp / Model Paper-II / University Exam 2026*

MIC23 AUTONOMOUS QUESTION BANK REPOSITORY

UNIT - II : PROCESSES, THREADS & CPU SCHEDULING

SOURCE VERIFICATION: Questions meticulously extracted from Official University Model Papers, Mid-1 / Mid-2 Question Banks, and Autonomous Semester End Examinations.

Regulation: MIC23 | Course Code: 23AM5T03 / 23CS4T01 | B.Tech V Semester (AI&ML / CSE)

UNIT - II - SECTION A: 2-MARK SHORT ANSWER QUESTIONS [2 MARKS EACH]

Q1. What are the different states of a process?
[2M] • *Source: University Exam 2026 / Mid-1 Imp*

Q2. Differentiate preemptive and non-preemptive scheduling.
[2M] • *Source: University Exam 2026 / Model Paper-I*

Q3. What is a Process Control Block (PCB)? What does PCB contain?
[2M] • *Source: Model Paper-I / Mid-1 Imp*

Q4. List any two CPU scheduling algorithms.
[2M] • *Source: Model Paper-I*

Q5. Differentiate between process and thread.
[2M] • *Source: Model Paper-II*

Q6. What is the role of a dispatcher in CPU scheduling?
[2M] • *Source: Model Paper-II / Model Paper-I*

Q7. Define process.
[2M] • *Source: Mid-1 Imp*

Q8. Define schedulers.
[2M] • *Source: Mid-1 Imp*

Q9. What are the types of schedulers?
[2M] • *Source: Mid-1 Imp*

Q10. Define Thread.
[2M] • *Source: Mid-1 Imp*

Q11. What is a multiprocessor system?
[2M] • *Source: Mid-1 Imp*

Q12. Define Context Switching.
[2M] • *Source: Mid-1 Imp*

Q13. List the Thread libraries.
[2M] • *Source: Mid-1 Imp*

UNIT - II: PROCESSES, THREADS & CPU SCHEDULING

SECTION B: 5-MARK LONG ANSWER QUESTIONS [5 MARKS EACH]

Q1. Describe message-passing and shared-memory models of Inter-Process Communication (IPC).
[5M] • *Source: University Exam 2026 / Mid-1 Imp*

Q2. Discuss multithreading models in detail with neat diagrams.
[5M] • *Source: University Exam 2026 / Model Paper-II / Mid-1 Imp*

Q3. Discuss Inter-Process Communication (IPC) and its primary mechanisms.
[5M] • *Source: University Exam 2026 / Model Paper-II / Mid-1 Imp*

Q4. Describe the role of a dispatcher in CPU scheduling and explain dispatch latency.
[5M] • *Source: Model Paper-I*

Q5. Explain First-Come, First-Served (FCFS) scheduling with an example and discuss the Convoy Effect.
[5M] • *Source: Model Paper-I*

Q6. Differentiate between preemptive and non-preemptive scheduling with suitable examples.
[5M] • *Source: Model Paper-I*

Q7. Write short notes on thread libraries (POSIX Pthreads, Win32, and Java Threads).
[5M] • *Source: Model Paper-I*

Q8. Explain the process concept with a neat diagram of process states.
[5M] • *Source: Model Paper-II / Mid-1 Imp*

Q9. What is a Process Control Block (PCB)? List and explain its components in detail.
[5M] • *Source: Model Paper-II*

Q10. Explain about process schedulers (Long-term, Short-term, and Medium-term schedulers).
[5M] • *Source: Mid-1 Imp*

Q11. Explain CPU Scheduling Algorithms with examples (FCFS, SJF, Priority, Round Robin).
[5M] • *Source: Mid-1 Imp*

Q12. Explain the Context Switching mechanism, steps involved, and overhead factors.
[5M] • *Source: Mid-1 Imp*

UNIT - II: PROCESSES, THREADS & CPU SCHEDULING

SECTION C: 10-MARK COMPREHENSIVE ESSAY QUESTIONS (LAQs) [10 MARKS EACH]

Q1. Explain the process concept, process life cycle state transitions, and the internal architecture of the Process Control Block (PCB) in detail with neat diagrams. Compare processes and threads.

[10M] • *Source: Model Paper-II / University Exam 2026 / Mid-1 Imp*

Q2. Explain CPU Scheduling Criteria, Objectives, and Algorithms in detail. Compare FCFS, SJF, SRTF, Priority, and Round Robin scheduling.

[10M] • *Source: Mid-1 Imp / Model Paper-I / University Exam 2026*

UNIT - II: PROCESSES, THREADS & CPU SCHEDULING

SECTION D: NUMERICAL & ANALYTICAL PROBLEMS [5 / 10 MARKS]

Problem 1: Problem 1: First-Come, First-Served (FCFS) Scheduling Analysis

Consider four processes P1, P2, P3, and P4 with given arrival and burst times. Schedule them using First-Come, First-Served (FCFS) scheduling and compute the Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), and Response Time (RT) for each process, along with their system-wide averages.

[5M] • *Source: University Examination April/May 2026*

Problem 2: Problem 2: Shortest Job First (SJF) Scheduling with Arrival Times

Consider five processes P1 to P5 with different arrival and burst times. Calculate the Average Turnaround Time (Avg TAT) and Average Waiting Time (Avg WT) using Non-Preemptive Shortest Job First (SJF) scheduling.

[5M] • *Source: Handwritten Student Notes (Problem 2) / Model Exam Paper*

Problem 3: Problem 3: Priority Scheduling Algorithm Analysis

Consider seven processes P1 to P7 with given priority numbers, arrival times, and burst times. Assuming lower priority numbers represent higher priority (1 = highest priority), schedule the processes using Non-Preemptive Priority Scheduling and calculate the Completion Time, Turnaround Time, Waiting Time, Response Time, and their averages.

[5M] • *Source: Handwritten Student Notes (Problem 3) / Autonomous Question Bank*

Problem 4: Problem 4: Round Robin (RR) CPU Scheduling (Time Quantum = 2s)

Consider six processes P1 to P6 arriving at different times. Schedule the processes using the Round Robin (RR) scheduling algorithm with a Time Quantum (q) of 2 seconds. Trace the ready queue state transitions, draw the Gantt chart, and compute the Average Completion Time, Average Turnaround Time, and Average Waiting Time.

[5M] • *Source: Handwritten Student Notes (Problem 4) / Autonomous Question Bank*

Problem 5: Problem 5: Shortest Remaining Time First (SRTF / Preemptive SJF)

Consider four processes P1, P2, P3, and P4 arriving at times 0, 1, 2, 3 ms respectively. Using the Preemptive Shortest Job First (SRTF) algorithm, draw the Gantt chart and compute the Average Waiting Time (AWT) and Average Turnaround Time (ATAT).

[5M] • *Source: Autonomous Mid-1 Examination / Master Question Bank*

MIC23 AUTONOMOUS QUESTION BANK REPOSITORY

UNIT - III : PROCESS SYNCHRONIZATION & DEADLOCKS

SOURCE VERIFICATION: Questions meticulously extracted from Official University Model Papers, Mid-1 / Mid-2 Question Banks, and Autonomous Semester End Examinations.

Regulation: MIC23 | Course Code: 23AM5T03 / 23CS4T01 | B.Tech V Semester (AI&ML / CSE)

UNIT - III - SECTION A: 2-MARK SHORT ANSWER QUESTIONS [2 MARKS EACH]

Q1. Define busy waiting.
[2M] • *Source: University Exam 2026 / Mid-1 Imp*

Q2. What is starvation?
[2M] • *Source: University Exam 2026*

Q3. State Peterson's solution for the critical section problem.
[2M] • *Source: Model Paper-I*

Q4. Define deadlock in operating systems.
[2M] • *Source: Model Paper-I / Mid-1 Imp*

Q5. What is a semaphore? / Define semaphores.
[2M] • *Source: Model Paper-II / Mid-1 Imp*

Q6. Mention two conditions necessary for deadlock to occur.
[2M] • *Source: Model Paper-II*

Q7. Define race condition.
[2M] • *Source: Mid-1 Imp*

Q8. What are the requirements that a solution to the critical section problem must satisfy?
[2M] • *Source: Mid-1 Imp*

Q9. Name the classic problems of synchronization.
[2M] • *Source: Mid-1 Imp*

Q10. What is the critical section problem?
[2M] • *Source: Mid-1 Imp*

Q11. Define spinlock.
[2M] • *Source: Mid-1 Imp*

Q12. What is Process Synchronization?
[2M] • *Source: Mid-1 Imp*

Q13. List the four necessary conditions for a deadlock to occur.
[2M] • *Source: Mid-2 QB*

Q14. Differentiate between deadlock prevention and deadlock avoidance.
[2M] • *Source: Mid-2 QB*

Q15. What is a wait-for graph and resource-allocation graph?
[2M] • *Source: Mid-2 QB*

Q16. What is a safe state in deadlock avoidance?

[2M] • *Source: Mid-2 QB*

UNIT - III: PROCESS SYNCHRONIZATION & DEADLOCKS

SECTION B: 5-MARK LONG ANSWER QUESTIONS [5 MARKS EACH]

Q1. Explain how the Dining Philosophers Problem can be solved using semaphores.
[5M] • *Source: University Exam 2026 / Model Paper-I / Model Paper-II / Mid-1 Imp*

Q2. Explain deadlock avoidance with an example of Banker's Algorithm.
[5M] • *Source: University Exam 2026 / Mid-2 QB*

Q3. Explain the Readers–Writers Problem using semaphores.
[5M] • *Source: University Exam 2026*

Q4. Discuss the necessary conditions for deadlock with examples.
[5M] • *Source: University Exam 2026 / Model Paper-I / Model Paper-II / Mid-2 QB*

Q5. Define semaphores and explain how they are used in process synchronization.
[5M] • *Source: Model Paper-I*

Q6. Write short notes on Peterson's solution for the critical section problem.
[5M] • *Source: Model Paper-I / Mid-1 Imp*

Q7. Explain the critical section problem with an example.
[5M] • *Source: Model Paper-II / Mid-1 Imp*

Q8. What are mutex locks? Explain their usage with an example.
[5M] • *Source: Model Paper-II*

Q9. What is a Semaphore? Explain the Producer-Consumer problem using semaphores.
[5M] • *Source: Mid-1 Imp*

Q10. What is a Monitor? Explain with an example how monitors achieve synchronization.
[5M] • *Source: Mid-1 Imp*

Q11. Explain deadlock prevention and discuss the methods used to prevent deadlocks.
[5M] • *Source: Mid-2 QB*

Q12. Explain deadlock detection for systems with single and multiple instances of resource types.
[5M] • *Source: Mid-2 QB*

Q13. Discuss the methods of recovery from deadlock, including process termination and resource preemption.
[5M] • *Source: Mid-2 QB*

Q14. Explain the different methods used for handling deadlocks, highlighting the role of each method.
[5M] • *Source: Mid-2 QB*

UNIT - III: PROCESS SYNCHRONIZATION & DEADLOCKS

SECTION C: 10-MARK COMPREHENSIVE ESSAY QUESTIONS (LAQs) [10 MARKS EACH]

Q1. Explain the classic problems of process synchronization (Bounded Buffer, Readers-Writers, and Dining Philosophers) in detail using semaphores and monitors. Provide neat sketches and comparative analysis.

[10M] • *Source: Model Paper-I / Model Paper-II / University Exam 2026 / Mid-1 Imp*

Q2. Explain Deadlock Characterization, Prevention, Avoidance (Banker's Algorithm), and Detection & Recovery in detail with neat diagrams and comparative analysis.

[10M] • *Source: Mid-2 QB / Model Paper-I / Model Paper-II / University Exam 2026*

UNIT - III: PROCESS SYNCHRONIZATION & DEADLOCKS

SECTION D: NUMERICAL & ANALYTICAL PROBLEMS [5 / 10 MARKS]

Problem 1: Deadlock Detection Algorithm Analysis (OS Units 3,4,5 Problems Bank)

Consider a system with 5 processes (P0 to P4) and 3 resource types (A, B, C). Given the system snapshot below, determine whether the system is in a deadlocked state using the Deadlock Detection Algorithm:

- Allocation Matrix:
  P0: [0, 1, 0], P1: [2, 0, 0], P2: [3, 0, 3], P3: [2, 1, 1], P4: [0, 0, 2]
- Request Matrix:
  P0: [0, 0, 0], P1: [2, 0, 2], P2: [0, 0, 0], P3: [1, 0, 0], P4: [0, 0, 2]
- Available Vector:
  [0, 0, 0]

Tasks:
(i) Trace the algorithm step-by-step.
(ii) Identify the complete resource allocation sequence.
(iii) Determine whether the system is deadlocked and list any deadlocked processes.

[5M] • *Source: OS Units 3, 4, 5 Problems Bank / Autonomous Mid-2 Examination*

Problem 2: Banker's Algorithm: Safety & Resource Request Analysis (Problems Bank)

Consider a system with 5 processes (P0 to P4) and 3 resource types (A, B, C). Total resource instances: A = 10, B = 5, C = 7. Given the system snapshot at time t0:

- Allocation Matrix:
  P0: [0, 1, 0], P1: [2, 0, 0], P2: [3, 0, 2], P3: [2, 1, 1], P4: [0, 0, 2]
- Max Demand Matrix:
  P0: [7, 5, 3], P1: [3, 2, 2], P2: [9, 0, 2], P3: [2, 2, 2], P4: [4, 3, 3]
- Available Vector: [3, 3, 2]

Tasks:
(i) Compute the content of the Need Matrix.
(ii) Determine if the system is in a Safe State and find the Safe Sequence.
(iii) If Process P1 requests [1, 0, 2], can this request be granted immediately?

[5M] • *Source: OS Units 3, 4, 5 Problems Bank / University Examination*

MIC23 AUTONOMOUS QUESTION BANK REPOSITORY

UNIT - IV : MEMORY MANAGEMENT STRATEGIES & MASS STORAGE

SOURCE VERIFICATION: Questions meticulously extracted from Official University Model Papers, Mid-1 / Mid-2 Question Banks, and Autonomous Semester End Examinations.

Regulation: MIC23 | Course Code: 23AM5T03 / 23CS4T01 | B.Tech V Semester (AI&ML / CSE)

UNIT - IV - SECTION A: 2-MARK SHORT ANSWER QUESTIONS [2 MARKS EACH]

Q1. Define dynamic linking.
[2M] • *Source: University Exam 2026*

Q2. What is a page fault?
[2M] • *Source: University Exam 2026*

Q3. What is paging in memory management?
[2M] • *Source: Model Paper-I*

Q4. Define demand paging.
[2M] • *Source: Model Paper-I / Mid-2 QB*

Q5. What is thrashing in operating systems?
[2M] • *Source: Model Paper-II / Mid-2 QB*

Q6. Define swapping in memory management.
[2M] • *Source: Model Paper-II*

Q7. Difference between internal fragmentation & external fragmentation.
[2M] • *Source: Mid-2 QB*

Q8. What is Translation Lookaside Buffer (TLB)?
[2M] • *Source: Mid-2 QB*

Q9. What is copy-on-write?
[2M] • *Source: Mid-2 QB*

Q10. What is RAID? State its main purpose.
[2M] • *Source: Mid-2 QB*

UNIT - IV: MEMORY MANAGEMENT STRATEGIES & MASS STORAGE

SECTION B: 5-MARK LONG ANSWER QUESTIONS [5 MARKS EACH]

Q1. Discuss demand paging and describe page fault handling.
[5M] • *Source: University Exam 2026 / Model Paper-I / Mid-2 QB*

Q2. Explain contiguous memory allocation and discuss fragmentation issues (internal and external).
[5M] • *Source: University Exam 2026 / Model Paper-I / Mid-2 QB*

Q3. What is Translation Lookaside Buffer (TLB)? Explain its working with a diagram.
[5M] • *Source: University Exam 2026*

Q4. Explain FCFS and SSTF disk scheduling algorithms with examples.
[5M] • *Source: University Exam 2026*

Q5. What is paging? Draw and explain the structure of a page table.
[5M] • *Source: Model Paper-I*

Q6. Differentiate between swapping and paging with suitable examples.
[5M] • *Source: Model Paper-I*

Q7. Define virtual memory. Explain the concept of demand paging.
[5M] • *Source: Model Paper-I*

Q8. What is copy-on-write? Explain its significance in memory management.
[5M] • *Source: Model Paper-II*

Q9. Explain the working of page replacement algorithms with an example of FIFO.
[5M] • *Source: Model Paper-II*

Q10. Define thrashing. Why does it occur, and how can it be controlled?
[5M] • *Source: Model Paper-II / Mid-2 QB*

Q11. Explain different HDD scheduling algorithms with an example (SCAN, C-SCAN, LOOK, C-LOOK).
[5M] • *Source: Model Paper-II*

Q12. Describe disk structure, disk attachment, and the major activities involved in disk management.
[5M] • *Source: Mid-2 QB*

UNIT - IV: MEMORY MANAGEMENT STRATEGIES & MASS STORAGE

SECTION C: 10-MARK COMPREHENSIVE ESSAY QUESTIONS (LAQs) [10 MARKS EACH]

Q1. Explain Paging Hardware, Address Translation, Translation Lookaside Buffer (TLB), and Demand Paging with Page Fault Handling in detail with neat diagrams and comparative analysis.

[10M] • *Source: Model Paper-I / Model Paper-II / University Exam 2026 / Mid-2 QB*

Q2. Explain Mass-Storage Structure, Hard Disk Geometry, Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK) in detail with comparative analysis and neat sketches.

[10M] • *Source: Mid-2 QB / Model Paper-I / Model Paper-II / University Exam 2026*

UNIT - IV: MEMORY MANAGEMENT STRATEGIES & MASS STORAGE

SECTION D: NUMERICAL & ANALYTICAL PROBLEMS [5 / 10 MARKS]

Problem 1: Problem 1: FIFO (First-In, First-Out) Page Replacement Algorithm

Consider the page reference string given below for a memory system with 3 physical frames (initially empty):

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Apply the FIFO page replacement algorithm. Construct the page replacement grid table showing the contents of all three frames after each reference, mark Hits (H) and Page Faults (PF), explain page replacements, and calculate the Total Hits, Total Page Faults, Hit Ratio, and Page Fault Ratio.

[5M] • *Source: Handwritten Student Notes / Autonomous Question Bank*

Problem 2: Problem 2: LRU (Least Recently Used) Page Replacement Algorithm

Using the same reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 with 3 frames, apply the Least Recently Used (LRU) algorithm. Construct the frame allocation grid table, explain replacement decisions based on past references, and compute Total Hits, Total Faults, Hit Ratio, and Fault Ratio.

[5M] • *Source: Handwritten Student Notes / Autonomous Question Bank*

Problem 3: Problem 3: Optimal (OPT) Page Replacement Algorithm

Apply the Optimal Page Replacement algorithm to the reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 with 3 frames. Draw the complete grid table, explain why each replacement guarantees the absolute minimum number of page faults, and compute the final Hit and Fault ratios.

[5M] • *Source: Handwritten Student Notes / Autonomous Question Bank*

Problem 4: Problem 4: Comparative Analysis of FIFO vs LRU vs Optimal

Summarize and compare the performance metrics, page fault counts, hit ratios, and architectural characteristics of FIFO, LRU, and Optimal page replacement algorithms on the 20-page reference workload.

[5M] • *Source: Autonomous Mid-2 Examination / Master Question Bank*

Problem 5: Problem 5: Comprehensive Disk Scheduling (FCFS, SSTF, SCAN, LOOK, C-SCAN, C-LOOK)

A disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53. The queue of pending requests, in FIFO order, is:

98, 183, 37, 122, 14, 124, 65, 67

Calculate the total head movement (in cylinders) for:
(i) FCFS
(ii) SSTF
(iii) SCAN (trace moving towards 0 first as in handwritten notes, and towards 199)
(iv) LOOK
(v) C-SCAN
(vi) C-LOOK

[5M] • *Source: Handwritten Student Notes (Problem 6) / Autonomous Mid-2 Examination*

Problem 6: Problem 6: Effective Memory Access Time (EMAT) with TLB Analysis

Consider a paging hardware architecture equipped with a Translation Lookaside Buffer (TLB). Assume that the entire page table and all process pages reside in physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6 (60%), calculate the Effective Memory Access Time (EMAT) with a complete step-by-step explanation.

[5M] • *Source: OS Units 3,4,5 Problems Bank / Autonomous University Examination*

Problem 7: Problem 7: Memory Partition Allocation Strategies (First-Fit, Best-Fit, Worst-Fit)

Given five memory partitions of 500 KB, 100 KB, 200 KB, 300 KB, and 600 KB (in order), determine how each of the First-Fit, Best-Fit, and Worst-Fit algorithms would place four processes requiring 212 KB, 417 KB, 112 KB, and 426 KB (in order). Which algorithm makes the most efficient use of memory?

[5M] • *Source: OS Units 3,4,5 Problems Bank / University Examination*

MIC23 AUTONOMOUS QUESTION BANK REPOSITORY

UNIT - V : FILE SYSTEM INTERFACE, IMPLEMENTATION & PROTECTION

SOURCE VERIFICATION: Questions meticulously extracted from Official University Model Papers, Mid-1 / Mid-2 Question Banks, and Autonomous Semester End Examinations.

Regulation: MIC23 | Course Code: 23AM5T03 / 23CS4T01 | B.Tech V Semester (AI&ML / CSE)

UNIT - V - SECTION A: 2-MARK SHORT ANSWER QUESTIONS [2 MARKS EACH]

Q1. List the attributes of a file.

[2M] • *Source: University Exam 2026*

Q2. Define free space management.

[2M] • *Source: University Exam 2026*

Q3. What is a directory structure in file systems?

[2M] • *Source: Model Paper-I*

Q4. State the goal of protection in an operating system.

[2M] • *Source: Model Paper-I / Mid-2 QB*

Q5. What is an access matrix in protection?

[2M] • *Source: Model Paper-II / Mid-2 QB*

Q6. Define file mounting.

[2M] • *Source: Model Paper-II*

Q7. List different file access methods.

[2M] • *Source: Mid-2 QB*

UNIT - V: FILE SYSTEM INTERFACE, IMPLEMENTATION & PROTECTION

SECTION B: 5-MARK LONG ANSWER QUESTIONS [5 MARKS EACH]

Q1. Discuss different access methods of a file system with examples.
[5M] • *Source: University Exam 2026 / Model Paper-I / Mid-2 QB*

Q2. Explain file system mounting and its significance.
[5M] • *Source: University Exam 2026*

Q3. Discuss file sharing mechanisms and protection issues.
[5M] • *Source: University Exam 2026*

Q4. Explain different allocation methods in file systems (contiguous, linked, indexed).
[5M] • *Source: University Exam 2026 / Model Paper-I / Mid-2 QB*

Q5. What is a file system? Explain different file access methods with examples.
[5M] • *Source: Model Paper-I*

Q6. Explain the various types of directory structures.
[5M] • *Source: Model Paper-I*

Q7. Describe the file-system operations performed by the operating system.
[5M] • *Source: Model Paper-I*

Q8. What is free space management? Explain any two methods with examples.
[5M] • *Source: Model Paper-II*

Q9. Explain the concepts of file-system mounting and partitions.
[5M] • *Source: Model Paper-II / Mid-2 QB*

Q10. Define protection in operating systems. Explain the principles of protection.
[5M] • *Source: Model Paper-II*

Q11. Explain the access matrix model for implementing protection in file systems.
[5M] • *Source: Model Paper-II / Mid-2 QB*

Q12. Explain protection rings and their role in operating-system protection.
[5M] • *Source: Mid-2 QB*

Q13. Explain directory implementation techniques and free-space management methods.
[5M] • *Source: Mid-2 QB*

UNIT - V: FILE SYSTEM INTERFACE, IMPLEMENTATION & PROTECTION

SECTION C: 10-MARK COMPREHENSIVE ESSAY QUESTIONS (LAQs) [10 MARKS EACH]

Q1. Explain File Allocation Methods (Contiguous, Linked, and Indexed Allocation) and Free-Space Management Techniques in detail with neat diagrams, multi-level indexing, and comparative analysis.

[10M] • *Source: Model Paper-I / Model Paper-II / University Exam 2026 / Mid-2 QB*

Q2. Explain Operating System Protection and Security, the Access Matrix Model, Implementation via Access Control Lists (ACL) and Capabilities, and Hardware Protection Rings in detail with neat sketches.

[10M] • *Source: Mid-2 QB / Model Paper-I / Model Paper-II / University Exam 2026*